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(54) **ION MILLING DEVICE, HOLDER, AND CROSS-SECTION MILLING TREATMENT METHOD**

(71) Applicant: **Hitachi High-Tech Corporation**, Minato-ku, Tokyo (JP)

(72) Inventors: **Saki FUKUSHIMA**, Tokyo (JP); **Masanari FURIKI**, Tokyo (JP); **Hisayuki TAKASU**, Tokyo (JP)

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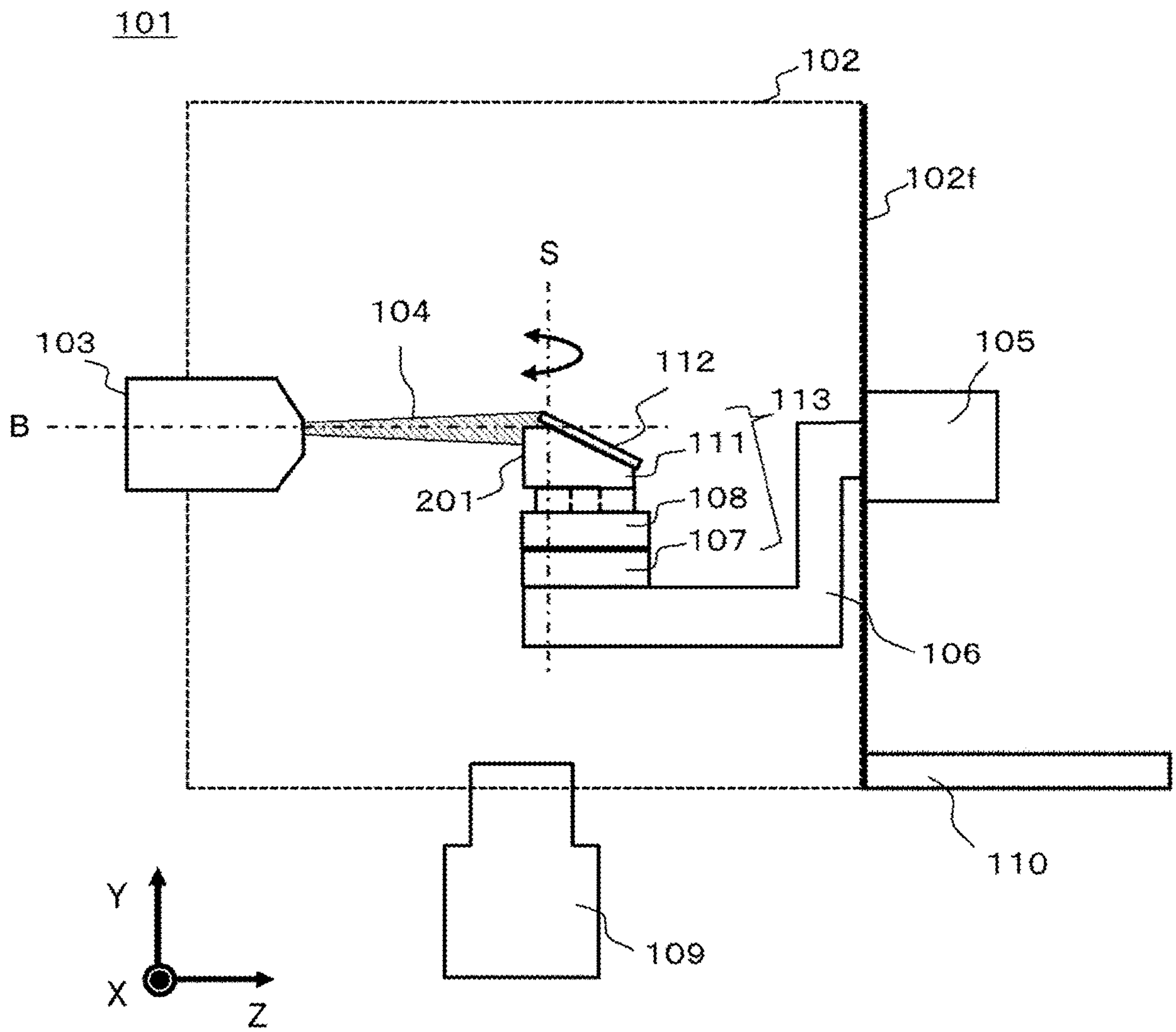
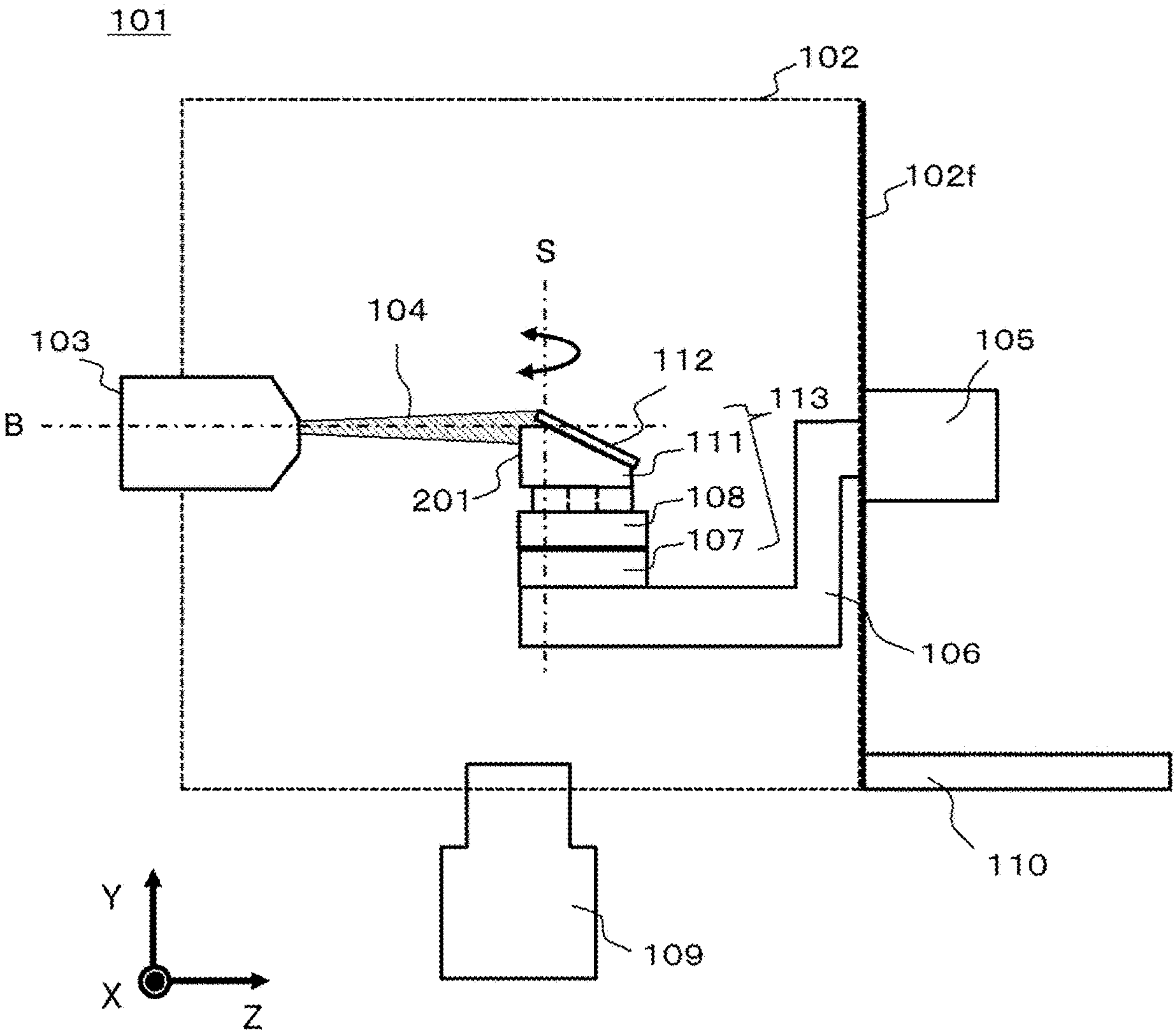
(57) **ABSTRACT**

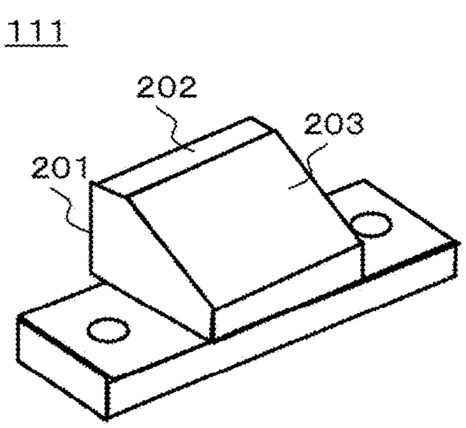
Provided is an ion milling device including a mask and sample stage unit **113** including a holder **111** to which a sample **112** adheres, a sample unit base **106** on which the mask and sample stage unit is mounted, and an ion source **103** that emits an unfocused ion beam **104** to the sample, in which the holder has a first surface to a third surface, a first surface **201** and a second surface **203** to which the sample adheres are connected by a third surface **202**, an angle formed between the first surface and the third surface is a right angle, an angle formed between the first surface and the second surface is an acute angle, and when the mask and sample stage unit is mounted on the sample unit base such that the mask and sample stage unit faces the ion source, the first surface of the holder is perpendicular to an ion beam center of the ion beam.



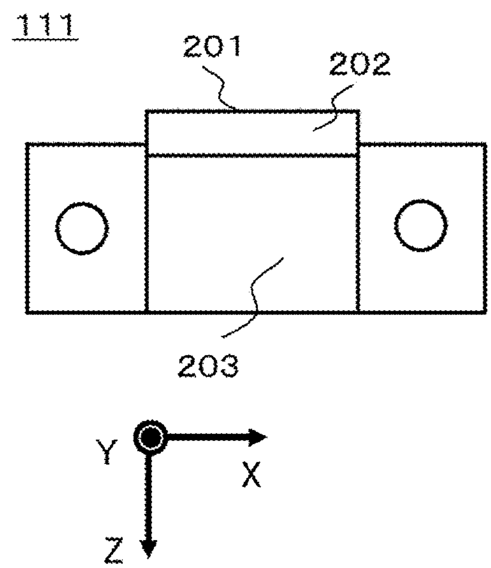
[FIG. 1]


101
102
102f
S
104
103
B
112
113
105
111
201
108
107
106
110
109
Y
X
Z

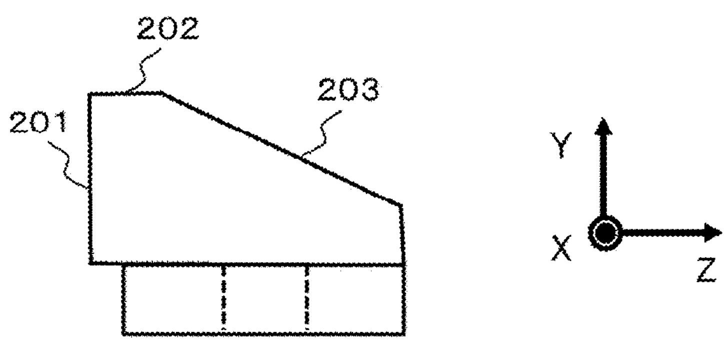
[FIG. 2A]


111
202
203
201

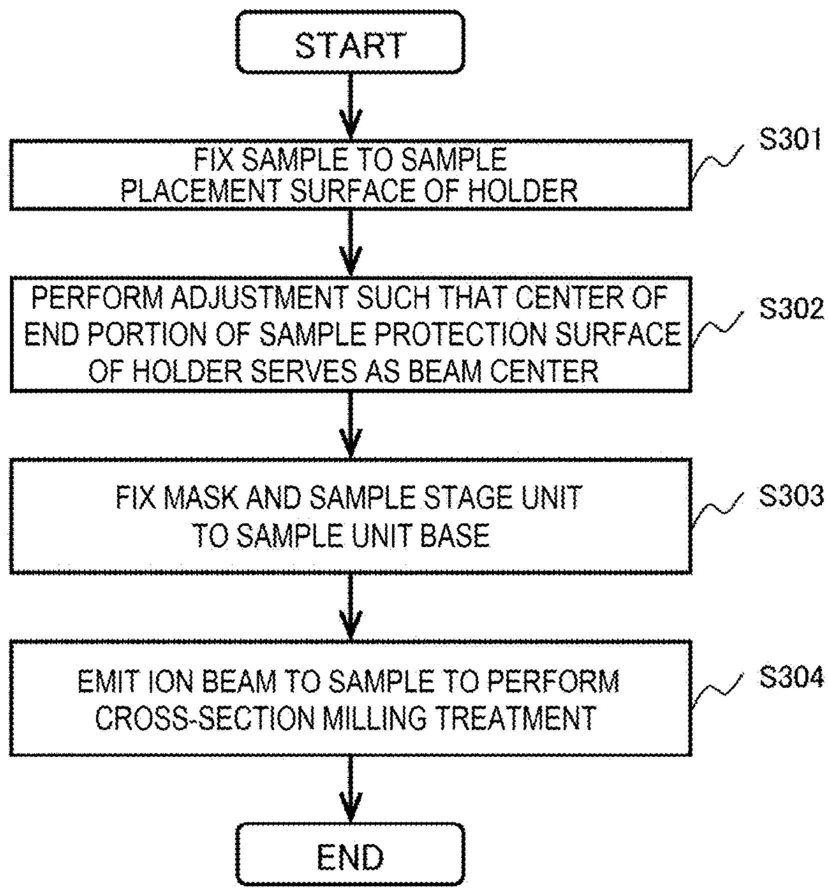
[FIG. 2B]



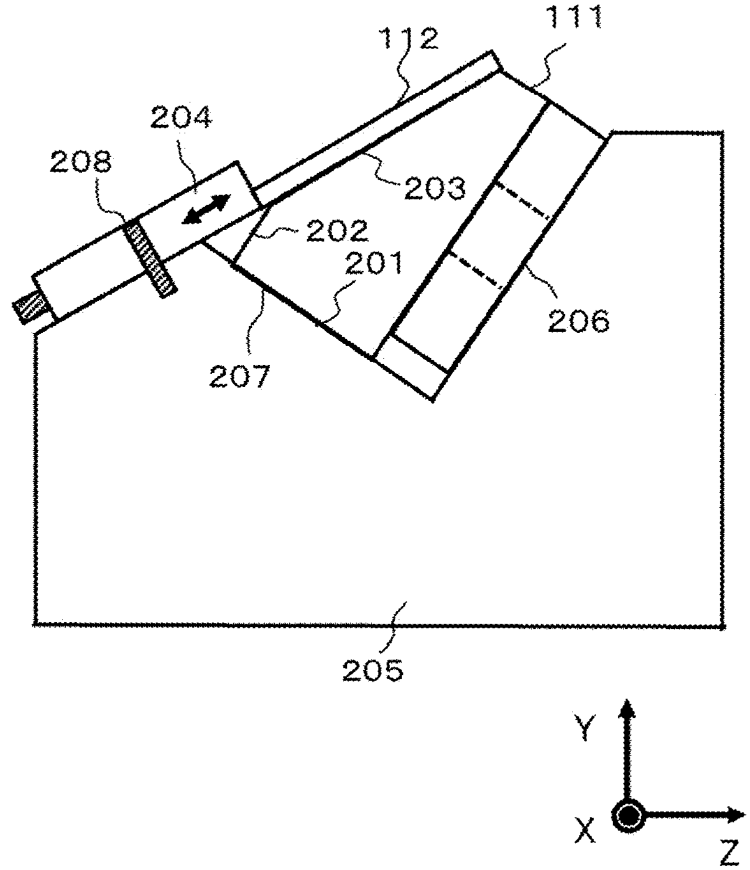
[FIG. 2C]



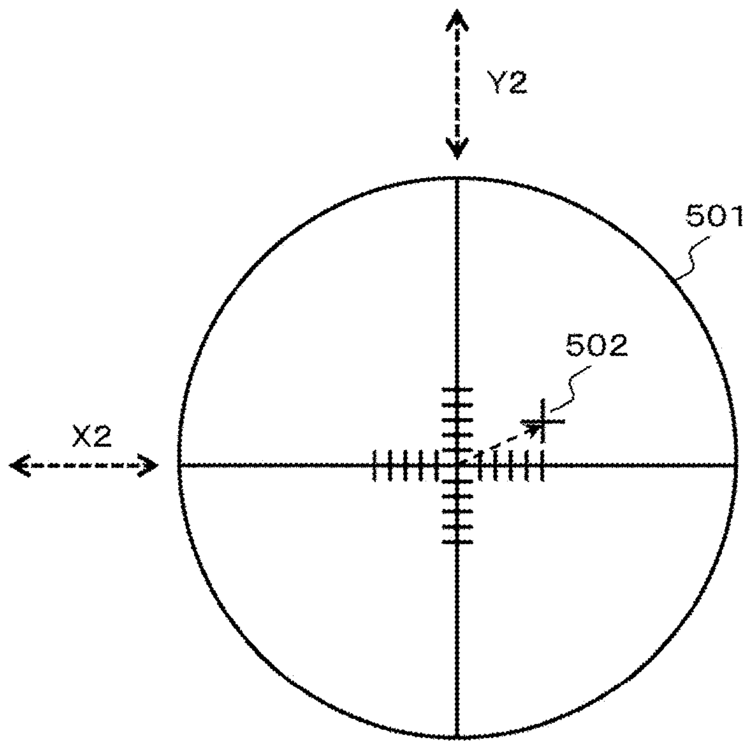
[FIG. 3]


START
FIX SAMPLE TO SAMPLE PLACEMENT SURFACE OF HOLDER
S301
PERFORM ADJUSTMENT SUCH THAT CENTER OF END PORTION OF SAMPLE PROTECTION SURFACE OF HOLDER SERVES AS BEAM CENTER
S302
FIX MASK AND SAMPLE STAGE UNIT TO SAMPLE UNIT BASE
S303
EMIT ION BEAM TO SAMPLE TO PERFORM CROSS-SECTION MILLING TREATMENT
S304
END

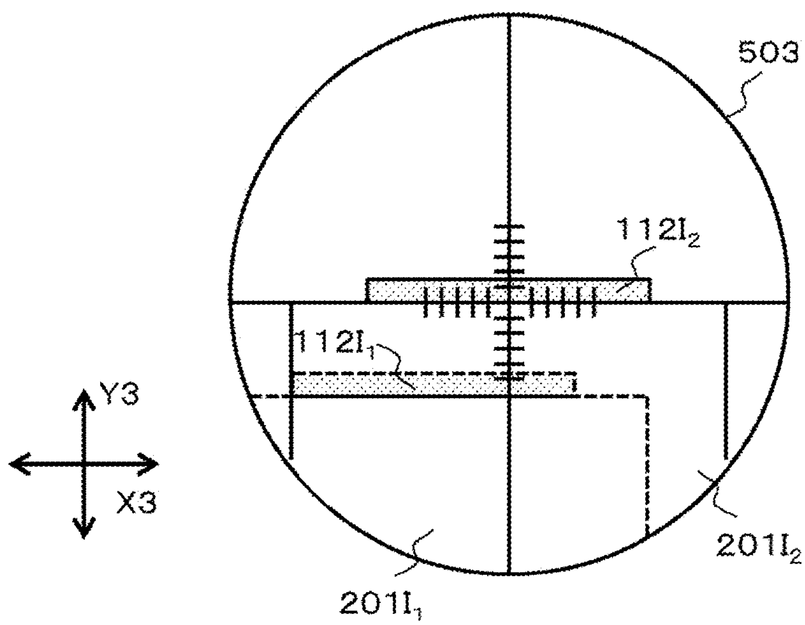
[FIG. 4]



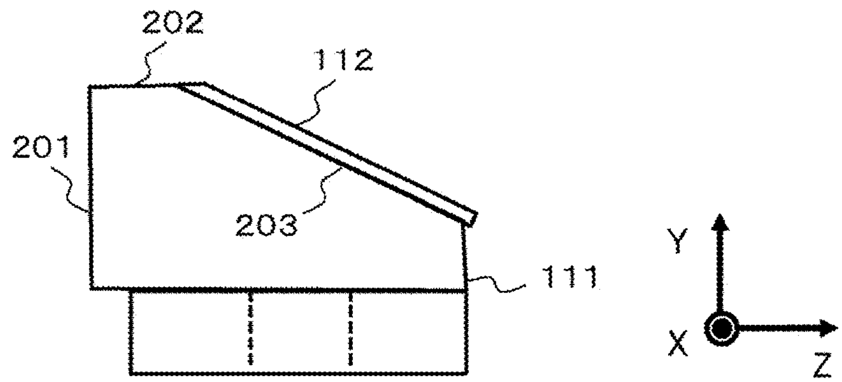
[FIG. 5A]



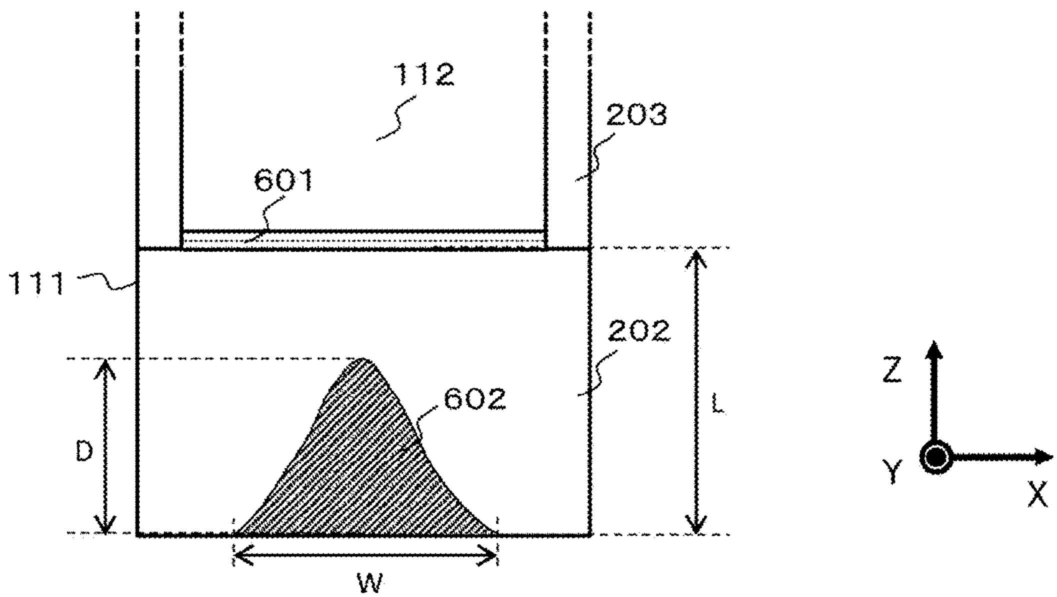
[FIG. 5B]



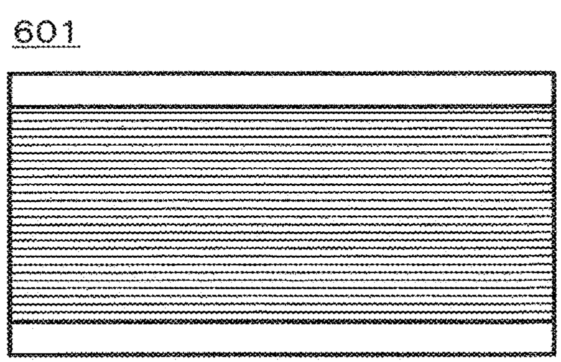
[FIG. 6A]



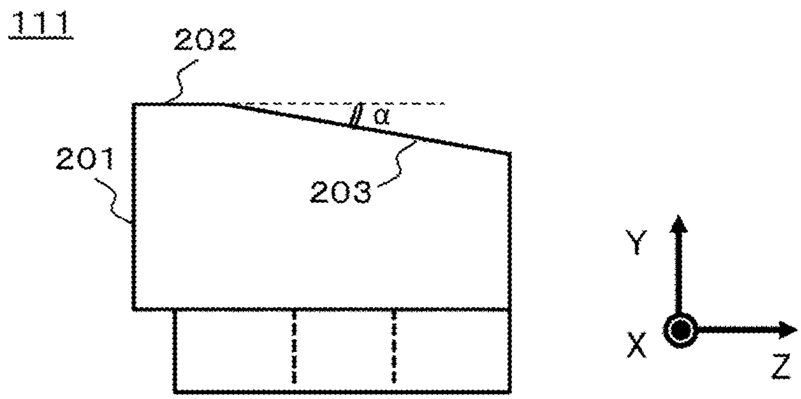
[FIG. 6B]



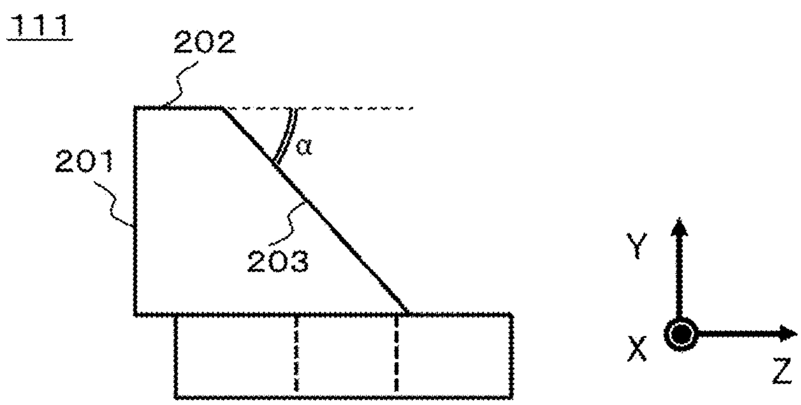
[FIG. 6C]



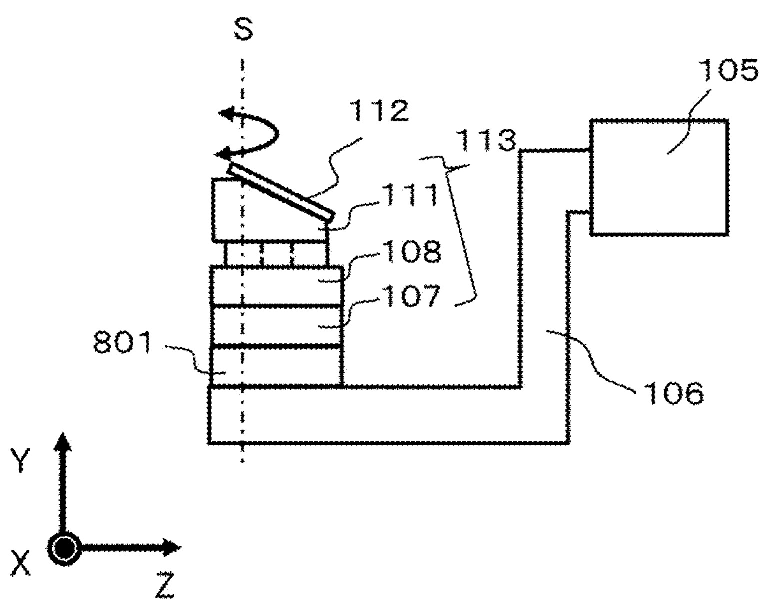
[FIG. 7A]



[FIG. 7B]



[FIG. 8]



ION MILLING DEVICE, HOLDER, AND CROSS-SECTION MILLING TREATMENT METHOD

TECHNICAL FIELD

[0001] The present invention relates to an ion milling device, a holder, and a cross-section milling treatment method.

BACKGROUND ART

[0002] An ion milling device for emitting an unfocused ion beam and exposing a cross-section to observe an internal structure of a sample has been known. PTL 1 discloses an ion milling device for performing cross-section milling in which a mask (shielding plate) disposed above a sample shields a part of an ion beam to expose a cross-section of the sample along an end surface of the mask. Further, in the ion milling device according to PTL 1, it is possible to slide a sample mask unit to form a cross-section having a width equal to or larger than an ion beam width or perform processing at a plurality of points.

[0003] On the other hand, a FIB-SEM for processing a sample by a focused ion beam (FIB) and performing observation and measurement by a scanning electron microscope has been known. PTL 2 discloses that an inclined cross-section of a sample including a circuit element is exposed by irradiation with a focused ion beam, and measurement of a pattern in a depth direction is performed.

CITATION LIST

Patent Literature

[0004] PTL 1: WO2017/145371

[0005] PTL 2: WO2016/002341

SUMMARY OF INVENTION

Technical Problem

[0006] When an inclined cross-section of a sample is exposed, it is difficult to obtain an inclined cross-section with a cut edge at a desired angle by controlling an inclination angle in the ion milling device according to PTL 1. On the other hand, in the focused ion beam device according to PTL 2, the processing of the inclined cross-section can be implemented only in a narrow region having a width and a depth of about several hundred micrometers.

[0007] An object of the invention is to provide an ion milling device capable of producing an inclined cross-section, which is precisely controlled over a wide area, of a laminated structure called a coating film or a functional thin film or a three-dimensional device such as MEMS.

Solution to Problem

[0008] An ion milling device according to an embodiment of the invention includes a mask and sample stage unit including a holder to which a sample adheres, a sample unit base on which the mask and sample stage unit is mounted, and an ion source configured to emit an unfocused ion beam to the sample. The holder has a first surface to a third surface, the first surface and the second surface to which the sample adheres are connected to each other by the third surface, an angle formed between the first surface and the third surface is a right angle, and an angle formed between the first surface and the second surface is an acute angle. When the mask and sample stage unit is mounted on the sample unit base such that the mask and sample stage unit faces the ion source, the first surface of the holder is perpendicular to an ion beam center of the ion beam.

Advantageous Effects of Invention

[0009] Provided is an ion milling device capable of producing an inclined cross-section precisely controlled. Other objects and novel features will become apparent from the description of this description and the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a diagram showing a configuration example of an ion milling device.

[0011] FIG. **2**A is a bird's-eye view of a holder.

[0012] FIG. **2**B is a top view of the holder.

[0013] FIG. **2**C is a side view of the holder.

[0014] FIG. **3** is an operation flow of performing a cross-section milling treatment for producing an inclined cross-section of a sample.

[0015] FIG. **4** is a diagram illustrating a method for adjusting a protrusion amount of a sample using an adjustment stage.

[0016] FIG. **5**A is a diagram illustrating a method for adjusting an irradiation position of an ion beam.

[0017] FIG. **5**B is a diagram illustrating the method for adjusting the irradiation position of the ion beam.

[0018] FIG. **6**A is a diagram showing states of the holder and the sample after the cross-section milling treatment is completed.

[0019] FIG. **6**B is a diagram showing the states of the holder and the sample after the cross-section milling treatment is completed.

[0020] FIG. **6**C is a schematic view of a sample cross-section.

[0021] FIG. **7**A shows an example of a holder having a sample placement surface with a small inclination angle.

[0022] FIG. **7**B shows an example of a holder having a sample placement surface with a large inclination angle.

[0023] FIG. **8** shows an example in which a mask and sample stage unit is mounted on a sample unit base **106** via a slide movement mechanism.

DESCRIPTION OF EMBODIMENTS

[0024] FIG. **1** shows a configuration example of an ion milling device **101**. The ion milling device **101** includes, as main components, a vacuum chamber **102**, an ion source **103** attached to the vacuum chamber **102**, a sample stage **105** attached to a surface of the vacuum chamber **102**, which is different from the surface to which the ion source **103** is attached, a sample unit base **106** extending from the sample stage **105**, a mask and sample stage unit **113** which is placed on the sample unit base **106** and on which a sample **112** to be subjected to a cross-section milling treatment is placed, a vacuum exhaust system **109** exhausting an inside of the vacuum chamber **102**, and a linear guide **110** provided on the surface of the vacuum chamber **102** to which the sample stage **105** is attached.

[0025] The sample stage **105** is attached to a flange **102***f* that also serves as a part of a container wall of the vacuum chamber **102**, and the vacuum chamber **102** can be opened to the atmosphere by pulling out the flange **102***f* along the linear guide **110**. At this time, the sample stage **105** is pulled out to the outside of the vacuum chamber **102** together with the sample unit base **106**. In this way, a sample stage extracting mechanism is configured.

[0026] The mask and sample stage unit **113** is assembled by stacking a micro-movement mechanism **107**, a coupling member **108**, and a holder **111** in this order, and the sample **112** is adhesively fixed onto the holder **111**. The holder **111** is an integrated mask and sample stage holder having a function as a sample stage on which the sample **112** is placed and a function as a mask for masking the sample **112**. The mask and sample stage unit **113** swings around a swing axis S perpendicular to a center (ion beam center) B of the ion beam **104** by a drive source provided in the sample stage **105**. In FIG. **1**, an ion beam center B is parallel to a Z-axis, the swing axis S is parallel to a Y-axis, the ion source **103** and the mask and sample stage unit **113** face each other, and a boundary (edge) between the holder **111** and the sample **112** is parallel to an X-axis. By the swing operation, the mask and sample stage unit **113** is rotationally driven in a range in which an angle formed between the edge and the X-axis direction is ±θ.

[0027] The micro-movement mechanism **107** constituting the mask and sample stage unit **113** is configured to be movable in a plane perpendicular to the ion beam center B, that is, in two axes including an X-axis direction and a Y-axis direction in the state shown in FIG. **1**, and is used to adjust a relative position between the ion beam center B and the mask and sample stage unit **113**. Details of the adjustment will be described below. The coupling member **108** serves as a base for placing the holder **111** on the micro-movement mechanism **107**.

[0028] FIGS. **2**A to **2**C show a shape of the holder **111**. FIG. **2**A is a bird's-eye view, FIG. **2**B is a top view, and FIG. **2**C is a side view. In FIGS. **2**B and **2**C, coordinate axes corresponding to the coordinate axes in FIG. **1** are displayed. The holder **111** includes a mask surface (first surface) **201** facing the ion source **103** and a sample placement surface (second surface) **203** to which the sample **112** is adhesively fixed when the mask and sample stage unit **113** faces the ion source **103**, and the mask surface **201** and the sample placement surface **203** are connected by a sample protection surface (third surface) **202**. The sample protection surface **202** is provided so that a portion of the sample **112** except for a target processing position (a position where a cross-section is desired to be exposed) is not scraped by the irradiation with an ion beam. In order to reduce the amount of the holder **111** scraped by the irradiation with the ion beam, a material of the holder **111** is preferably titanium, graphite carbon, or the like having high ion beam resistance.

[0029] The mask surface **201** is a plane parallel to an XY plane, the sample protection surface **202** is a plane parallel to an XZ plane, and therefore, an angle between the mask surface **201** and the sample protection surface **202** is a right angle. The sample placement surface **203** is inclined in the Z direction, and an angle formed between the mask surface **201** and the sample placement surface **203** is an acute angle. An intersection line between the mask surface **201** and the sample protection surface **202** is parallel to an intersection line between the sample protection surface **202** and the sample placement surface **203**.

[0030] A procedure for producing the inclined cross-section of the sample **112** placed on the mask and sample stage unit **113** by the ion milling device **101** will be described with reference to FIG. **3**.

[0031] S**301**: The user adhesively fixes the sample **112** to the sample placement surface **203** of the holder **111**. As shown in FIG. **1**, a portion of the sample **112** protruding from the intersection line between the sample protection surface **202** and the sample placement surface **203** of the holder **111** is a portion to be cross-section milled by the ion beam **104**. Therefore, when the sample **112** is fixed to the holder **111**, the protrusion amount is adjusted using an adjustment stage **205** so that the target processing position of the sample **112** is positioned on the intersection line between the sample protection surface **202** and the sample placement surface **203**. FIG. **4** shows a state in which the protrusion amount of the sample **112** is adjusted by placing the holder **111** on the adjustment stage **205**.

[0032] The adjustment stage **205** is a stage for adjusting the protrusion amount of the sample **112**, which is produced in accordance with the holder **111**. The adjustment stage **205** is provided with a triangular groove on which the holder **111** is mounted. The adjustment of the protrusion amount is performed in a state where the holder **111** is placed on the adjustment stage **205** such that a bottom surface of the holder **111** comes into contact with a placement surface **206** which is one side surface of the triangular groove, and the mask surface **201** of the holder **111** comes into contact with an adjustment surface **207** which is the other side surface of the triangular groove.

[0033] In order to precisely adjust the protrusion amount, the adjustment stage **205** is provided with a protrusion amount adjustment jig **204**. The triangular groove is formed at an inclination such that when the sample **112** is placed on the sample placement surface **203** of the holder **111**, the sample **112** slides down toward the sample protection surface **202** by its own weight. The sample **112** is held on the sample placement surface **203** by the protrusion amount adjustment jig **204**, and the sample protrusion amount can be set by sliding a protrusion amount adjustment bar **208** in arrow directions. The protrusion amount of the sample can be precisely adjusted by using a precise screw mechanism used for a micrometer for the protrusion amount adjustment bar **208**. When the desired protrusion amount is reached, the sample **112** is adhesively fixed to the sample placement surface **203**.

[0034] The protrusion amount is set based on a distance from a distal end of the trimmed sample **112** to the target processing position when the sample **112** is trimmed. When the target processing position is not specified and processing is to be performed at any location, it is preferable to set the protrusion amount to about 50 μm. For fixing the sample **112** to the sample placement surface **203**, an adhesive such as a carbon paste, hot wax, or manicure can be used. When the sample **112** is fixed, the holder **111** is removed from the adjustment stage **205** and attached to the micro-movement mechanism **107** via the coupling member **108**.

[0035] S**302**: The ion beam center B of the ion beam **104** emitted from the ion source **103** is adjusted to be positioned at a center of an end portion of the sample protection surface **202** of the holder **111**. An example of the adjustment method will be described with reference to FIGS. **5**A and **5**B.

[0036] Since the relative position between the ion beam center B and the mask and sample stage unit **113** needs to be precisely adjusted, the adjustment is performed using an optical microscope. The optical microscope includes a fixing stage for reproducibly positioning the mask and sample stage unit **113** at a fixed position on an observation stage of the optical microscope. Prior to the adjustment in step S**302**, the field of view of the optical microscope is adjusted so that the ion beam center B is positioned at a center of the field of view of the optical microscope. This adjustment is performed at a timing of maintenance, and does not need to be performed every time the sample is placed on the holder.

[0037] The adjustment of the field of view of the optical microscope will be described with reference to FIG. **5**A. The micro-movement mechanism **107** is set at a standard position (X-axis direction=0, Y-axis direction=0), photosensitive paper, silver foil, or the like is attached to the mask surface **201** of the holder **111**, and the field of view of the optical microscope is moved so that a mark formed by emitting the ion beam **104**, that is, an ion beam center **502** is positioned at a center of a field of view **501** of the optical microscope.

[0038] FIG. **5**B shows a field of view **503** of the optical microscope when the mask and sample stage unit **113** on which the sample **112** is placed is disposed on the fixing stage of the optical microscope in step S**302**. When the micro-movement mechanism **107** is at the standard position (X-axis direction=0, Y-axis direction=0), an image of the sample **112** is a sample image **112**I_1, and an image of the mask surface **201** is a mask surface image **201**I_1. Since the ion beam center is the center of the field of view **503**, a position X**3** in the X-axis direction and a position Y**3** in the Y-axis direction of the micro-movement mechanism **107** may be adjusted so that the center of the end portion of the sample protection surface **202** of the holder **111** is positioned at the center of the field of view **503** according to the target processing position. Accordingly, the sample protection surface **202** of the holder is adjusted to be positioned along the ion beam center of the ion beam. An image of the sample **112** after adjustment by the micro-movement mechanism **107** is a sample image **112**I_2, and an image of the mask surface **201** is a mask surface image **201**I_2.

[0039] S**303**: The mask and sample stage unit **113** is attached to the sample unit base **106**. The attaching position of the sample unit base **106** to the mask and sample stage unit is fixed. As shown in FIG. **1**, the mask surface **201** of the holder **111** is fixed facing the ion source **103**.

[0040] S**304**: The ion beam **104** is emitted to the sample **112** to perform a cross-section milling treatment. When the irradiation direction of the ion beam **104** is concentrated in one direction, a processing mark appears in the cross-section of the sample. In order to prevent the occurrence of such disturbance of the cross-section, swing processing in which milling treatment is performed while swinging the mask and sample stage unit **113** around the swing axis S is performed.

[0041] As described above, a protruding region of the sample **112** adhesively fixed to the sample placement surface **203** is milled by irradiation with the ion beam.

[0042] FIGS. **6**A to **6**C show states of the holder **111** and the sample **112** after the cross-section milling treatment is completed. As shown in FIG. **6**A, a cross-section **601** of the sample **112** is exposed along the sample protection surface **202** of the holder **111**. The cross-section **601** is an inclined cross-section in which a cut edge is inclined in accordance with the inclination of the sample placement surface **203** of the holder **111**.

[0043] As shown in FIG. **1**, since the ion beam **104** is adjusted to be emitted such that the ion beam center B moves along the sample protection surface **202** during the milling treatment, the sample protection surface **202** of the holder **111** is also scraped together with the sample **112** by the irradiation with the ion beam **104**. This state is shown in FIG. **6**B. At this time, regarding a milling region **602** of the sample protection surface **202**, a length cut in an extension direction (Z-axis direction) of the ion beam center B is defined as a processing depth D, and a length of the milling region **602** on the end portion of the sample protection surface **202** is defined as a processing width W.

[0044] A length L of the sample protection surface **202** in the Z-axis direction affects the accuracy of milling. When the length L is too small, for example, L<0.1 mm, the milling region **602** reaches the sample placement surface **203** before completion of processing, and a cross-section along the sample protection surface **202** cannot be obtained. In contrast, when the length L is too large, for example, L>10 mm, a position of the sample **112** relative to the ion source **103** becomes farther, and thus the intensity of the ion beam **104** which is unfocused decreases. Accordingly, the processing rate decreases. When the processing time is extended due to the decrease in the processing rate, the influence of heating due to sputtering caused by the ion beam **104** on the sample **112** and the holder **111** is remarkable. Specifically, when a thermal expansion coefficient of the sample **112** is different from a thermal expansion coefficient of the holder **111**, the degree of adhesion between the holder **111** and the sample **112** decreases during the milling, and the problem that the sputtered particles enter the gap occurs. As a result, the processing accuracy decreases, that is, the cross-section **601** is curved. In order to avoid such a problem, the length L of the sample protection surface **202** in the Z-axis direction is 0.1 mm or more and 10 mm or less, preferably 0.5 mm or more and 3 mm or less, and particularly preferably 2 mm or less.

[0045] FIG. **6**C schematically shows a state of a cross-section **601** of the obtained sample **112** observed with a scanning electron microscope. The sample **112** has a structure in which a semi-conductor layer and an insulating film layer are laminated on a semi-conductor substrate. When the cross-section **601** is obtained as a flat surface without being curved, a boundary between parallel laminated films without distortion can be observed.

[0046] An inclination angle α of the sample placement surface **203** of the holder **111** can be changed by replacing the holder **111** in accordance with the sample **112** to be placed, and can be customized in a range of several degrees to 70 degrees or less. FIG. **7**A shows the holder **111** with the inclination angle α=10°, and FIG. **7**B shows the holder **111** with the inclination angle α=70°. Only the inclination angle α is changed, and the other sizes are the same. The holder **111** having the optimum inclination angle α can be produced in accordance with the type of the sample **112** to be placed and the observation/measurement purpose.

[0047] FIG. **8** shows an example in which the mask and sample stage unit **113** is mounted on the sample unit base **106** via a slide movement mechanism **801**. The slide movement mechanism **801** is a mechanism for implementing a wide-area milling treatment or a multi-point milling treatment. The slide movement mechanism **801** includes a drive source and reciprocates the mask and sample stage unit **113** in a direction of the intersection line between the sample protection surface **202** and the sample placement surface **203** of the holder **111**, here, in the X-axis direction. The wide-area milling treatment is a treatment of processing a region having a width larger than a width of the ion beam **104**, and the multi-point milling treatment is a treatment of processing a plurality of target processing positions of the sample.

[0048] When the wide-area milling treatment is performed, the milling treatment may be performed while performing both the reciprocating operation by the slide movement mechanism **801** and the swing operation around the swing axis S. When performing the multi-point milling treatment, the milling treatment may be performed at a plurality of target processing positions by moving the irradiation position of the ion beam **104** by the slide movement mechanism **801**.

[0049] The invention is not limited to the embodiments described above, and includes various modifications. For example, the above-described embodiments have been described in detail to facilitate understanding of the invention, and the invention is not necessarily limited to those including all the configurations described. A part of a configuration of a certain embodiment can be replaced with a configuration of another embodiment, and a configuration of another embodiment can be added to a configuration of a certain embodiment. A part of a configuration according to each embodiment may be added to, deleted from, or replaced with another configuration.

REFERENCE SIGNS LIST

[0050] **101**: ion milling device
[0051] **102**: vacuum chamber
[0052] **102*f***: flange
[0053] **103**: ion source
[0054] **104**: ion beam
[0055] **105**: sample stage
[0056] **106**: sample unit base
[0057] **107**: micro-movement mechanism
[0058] **108**: coupling member
[0059] **109**: vacuum exhaust system
[0060] **110**: linear guide
[0061] **111**: holder
[0062] **112**: sample
[0063] **113**: mask and sample stage unit
[0064] **201**: mask surface
[0065] **202**: sample protection surface
[0066] **203**: sample placement surface
[0067] **204**: protrusion amount adjustment jig
[0068] **205**: adjustment stage
[0069] **206**: placement surface
[0070] **207**: adjustment surface
[0071] **208**: protrusion amount adjustment bar
[0072] **501**, **503**: field of view
[0073] **502**: ion beam center
[0074] **601**: cross-section
[0075] **602**: milling region
[0076] **801**: slide movement mechanism

1. An ion milling device comprising:
a mask and sample stage unit including a holder to which a sample adheres;
a sample unit base on which the mask and sample stage unit is mounted; and
an ion source configured to emit an unfocused ion beam to the sample, wherein
the holder has a first surface to a third surface, the first surface and the second surface to which the sample adheres are connected to each other by the third surface, an angle formed between the first surface and the third surface is a right angle, an angle formed between the first surface and the second surface is an acute angle, and
when the mask and sample stage unit is mounted on the sample unit base such that the mask and sample stage unit faces the ion source, the first surface of the holder is perpendicular to an ion beam center of the ion beam.

2. The ion milling device according to claim **1**, wherein
the sample adheres to the second surface such that a protrusion amount from an intersection line between the second surface and the third surface of the holder reaches a predetermined protrusion amount.

3. The ion milling device according to claim **1**, wherein
a distance between an intersection line between the first surface and the third surface of the holder and an intersection line between the second surface and the third surface is 0.1 mm or more and 10 mm or less.

4. The ion milling device according to claim **1**, wherein
the mask and sample stage unit includes a micro-movement mechanism,
the micro-movement mechanism is configured to move the sample unit base in a plane perpendicular to the ion beam center of the ion beam in a state where the mask and sample stage unit faces the ion source, and
the micro-movement mechanism is adjusted to allow the third surface of the holder to be positioned along the ion beam center of the ion beam.

5. The ion milling device according to claim **1**, further comprising:
a sample stage configured to rotationally drive the mask and sample stage unit via the sample unit base, wherein
the sample stage swings the mask and sample stage unit around a swing axis perpendicular to the ion beam center of the ion beam and an intersection line between the second surface and the third surface of the holder.

6. The ion milling device according to claim **5**, wherein
the mask and sample stage unit is mounted on the sample unit base via a slide movement mechanism, and
the slide movement mechanism reciprocates the mask and sample stage unit in a direction of the intersection line between the second surface and the third surface of the holder.

7. A holder to which a sample to be cross-section milled by an unfocused ion beam adheres, the holder comprising:
a first surface to a third surface, wherein
the first surface and the second surface to which the sample adheres are connected to each other by the third surface, an angle formed between the first surface and the third surface is a right angle, an angle formed between the first surface and the second surface is an acute angle, and
the ion beam is emitted to the sample from a direction of the first surface.

8. The holder according to claim **7**, wherein
the sample adheres to the second surface to reach a predetermined protrusion amount from an intersection line between the second surface and the third surface of the holder.

9. The holder according to claim **7**, wherein

a distance between an intersection line between the first surface and the third surface and an intersection line between the second surface and the third surface is 0.1 mm or more and 10 mm or less.

10. The holder according to claim **7**, wherein

a material of the holder is titanium or graphite carbon.

11. A cross-section milling treatment method using an ion milling device including a mask and sample stage unit including a holder, a sample unit base on which the mask and sample stage unit is mounted, and an ion source, wherein

the holder has a first surface to a third surface, the first surface and the second surface are connected to each other by the third surface, an angle formed between the first surface and the third surface is a right angle, an angle formed between the first surface and the second surface is an acute angle,

a sample adheres to the second surface of the holder such that a protrusion amount from an intersection line between the second surface and the third surface of the holder reaches a predetermined protrusion amount, and

an unfocused ion beam is emitted from the ion source to the sample from a direction of the first surface of the holder.

12. The cross-section milling treatment method according to claim **11**, wherein

the mask and sample stage unit further includes a micro-movement mechanism,

the micro-movement mechanism is configured to move the sample unit base in a plane perpendicular to an ion beam center of the ion beam in a state where the mask and sample stage unit faces the ion source, and

the micro-movement mechanism is adjusted to allow the third surface of the holder to be positioned along the ion beam center of the ion beam.

13. The cross-section milling treatment method according to claim **11**, wherein

the ion milling device further includes a sample stage configured to rotationally drive the mask and sample stage unit via the sample unit base, and

the sample stage swings the mask and sample stage unit around a swing axis perpendicular to an ion beam center of the ion beam and an intersection line between the second surface and the third surface of the holder during a period in which the ion source emits the ion beam to the sample.

14. The cross-section milling treatment method according to claim **13**, wherein

the mask and sample stage unit is mounted on the sample unit base via a slide movement mechanism, and

the slide movement mechanism reciprocates the mask and sample stage unit in a direction of the intersection line between the second surface and the third surface of the holder during the period in which the ion source emits the ion beam to the sample.

15. The cross-section milling treatment method according to claim **14**, wherein

a region of the sample having a width wider than a width of the ion beam is processed or a plurality of target processing positions of the sample are processed.

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